



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Department of Artificial Intelligence

Indian Institute of Technology Hyderabad

Mid-Term Examination

AI3000: Reinforcement Learning

July-August 2026

Student Details

Name

Roll Number

16 September 2026

15 Marks

Part 01

Instructions

1. Fill in your name and roll number in the space provided above.
2. This part is for 15 marks.
The complete examination is for 30 marks and counts towards 30% of the total grade.
3. Write all your answers in the question paper, and return the question paper at the end of the exam.
4. Show all your working clearly. Intermediate logical steps may receive partial credit.
5. Plagiarism will not be entertained. If you are caught cheating during the examination, your answer script will not be evaluated.

Useful Results

Value Iteration (Span-based stopping criterion)

1. **Initialize:**

- (a) $v'(s)$ for $s \in \mathcal{S}$
- (b) $\varepsilon > 0$ and $\sigma > (1 - \lambda)\varepsilon/\lambda$

2. **Iterate:** While $\sigma > (1 - \lambda)\varepsilon/\lambda$

- (a) $\mathbf{v} \leftarrow \mathbf{v}'$
- (b) $\mathbf{v}' \leftarrow L\mathbf{v}$
- (c) $\sigma \leftarrow \text{sp}(\mathbf{v}' - \mathbf{v})$

3. **Terminate:** Return

- (a) $\phi_\varepsilon: \mathcal{S} \rightarrow \mathcal{A}$ defined via

$$\phi_\varepsilon(s) = \arg \max_{a \in \mathcal{A}} r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s' | s, a) v(s'), \quad s \in \mathcal{S}.$$

- (b) $v_{\infty, \lambda, \varepsilon}: \mathcal{S} \rightarrow \mathbb{R}$ defined via

$$v_{\infty, \lambda, \varepsilon}(s) = v'(s) + \frac{\lambda}{1 - \lambda} \min(\mathbf{v}' - \mathbf{v}), \quad s \in \mathcal{S},$$

where $\min(\mathbf{u}) = \min_{s \in \mathcal{S}} u(s)$.

Policy Iteration

1. **Initialize:**

- (a) $\phi: \mathcal{S} \rightarrow \mathcal{A}$
- (b) $\Gamma \leftarrow \mathcal{S}$

2. **Iterate:** While Γ is non-empty

- (a) Policy evaluation: Solve

$$\mathbf{v} = \mathbf{r}_\phi + \lambda P_\phi \mathbf{v}.$$

- (b) Policy Improvement: Set $\phi': \mathcal{S} \rightarrow \mathcal{A}$ as

$$\phi'(s) \in \arg \max_{a \in \mathcal{A}} r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s' | s, a) v(s'), \quad s \in \mathcal{S},$$

choosing $\phi'(s) = \phi(s)$ if $\phi(s)$ is an element of $\arg \max$.

- (c) $\Gamma \leftarrow \{s \in \mathcal{S} \mid \phi'(s) \neq \phi(s)\}$
- (d) $\phi \leftarrow \phi'$

3. **Terminate:** Return ϕ and $\mathbf{v}$.

Question 1 (2+2+2 Marks)

Consider a time-homogeneous MDP over finite state and action spaces, with a policy π , and stationary deterministic rewards that depend only on the previous state $R_{n+1} = r(S_n)$. Prove or disprove (by providing a counter-example) the following statements:

- (a) If $\pi \in \Pi^{\text{HR}} \setminus \Pi^{\text{MR}}$ then the state-action process $\{(S_n, A_n)\}_{n \geq 0}$ is not a Markov chain.
- (b) If $\pi \in \Pi^{\text{HR}} \setminus \Pi^{\text{MR}}$ then the state process $\{S_n\}_{n \geq 0}$ is a Markov chain.
- (c) If $\pi \in \Pi^{\text{MR}}$ then the reward process $\{R_n\}_{n \geq 1}$ is a Markov chain.

Question 2 (3+2 Marks)

Consider an infinite-horizon MDP with discount factor $\lambda = 0.5$, state space $\mathcal{S} = \{0, 1\}$, action space $\mathcal{A} = \{\alpha, \beta\}$, time-homogeneous TPM

$$P(s' | s, a) = \begin{cases} 0.8, & s' = 1 - s, a = \alpha, \\ 0.2, & s' = s, a = \alpha, \\ 0.2, & s' = 1 - s, a = \beta, \\ 0.8, & s' = s, a = \beta, \end{cases}$$

and deterministic rewards

$$r(s, a, s') = \begin{cases} \text{XOR}(s, s'), & a = \alpha, \\ \text{AND}(s, s'), & a = \beta. \end{cases}$$

Starting with the deterministic decision rule $\phi(0) = \beta, \phi(1) = \beta$, carry out policy iteration until convergence.

- (a) What is the optimal decision rule $\phi^*: \mathcal{S} \rightarrow \mathcal{A}$ returned by policy iteration?
- (b) Characterise all the optimal stationary deterministic decision rules for the above MDP.

Question 3 (4 Marks)

Consider an infinite horizon MDP with discount factor $\lambda < 1$ with state space $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$, where $\mathcal{S}_1 \cap \mathcal{S}_2 = \emptyset$, and finite action space $\mathcal{A}$. Suppose we have two distinct decision rules $\phi_1, \phi_2 : \mathcal{S} \rightarrow \mathcal{A}$ such that $\pi^{(1)} = \{\phi_1, \phi_1, \dots\}$ and $\pi^{(2)} = \{\phi_2, \phi_2, \dots\}$ are optimal stationary deterministic policies for the MDP. Define a new decision rule $\phi : \mathcal{S} \rightarrow \mathcal{A}$ via

$$\phi(s) = \begin{cases} \phi_1(s), & s \in \mathcal{S}_1, \\ \phi_2(s), & s \in \mathcal{S}_2. \end{cases}$$

Prove that $\pi = \{\phi, \phi, \dots\}$ is optimal.

Hint: First, show that $v_{\infty, \lambda}^* = L_{\phi} v_{\infty, \lambda}^*$, and subsequently justify that the policy $(\phi, \phi, \dots)$ is optimal.



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Part 02

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Useful Results

Value Iteration (Span-based stopping criterion)

1. **Initialize:**

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2. **Iterate:** While $\sigma > (1 - \lambda)\varepsilon/\lambda$

- (a) $\mathbf{v} \leftarrow \mathbf{v}'$
- (b) $\mathbf{v}' \leftarrow L\mathbf{v}$
- (c) $\sigma \leftarrow \text{sp}(\mathbf{v}' - \mathbf{v})$

3. **Terminate:** Return

- (a) $\phi_\varepsilon: \mathcal{S} \rightarrow \mathcal{A}$ defined via

$$\phi_\varepsilon(s) = \arg \max_{a \in \mathcal{A}} r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s' | s, a) v(s'), \quad s \in \mathcal{S}.$$

- (b) $v_{\infty, \lambda, \varepsilon}: \mathcal{S} \rightarrow \mathbb{R}$ defined via

$$v_{\infty, \lambda, \varepsilon}(s) = v'(s) + \frac{\lambda}{1 - \lambda} \min(\mathbf{v}' - \mathbf{v}), \quad s \in \mathcal{S},$$

$$\text{where } \min(\mathbf{u}) = \min_{s \in \mathcal{S}} u(s).$$

Policy Iteration

1. **Initialize:**

- (a) $\phi: \mathcal{S} \rightarrow \mathcal{A}$
- (b) $\Gamma \leftarrow \mathcal{S}$

2. **Iterate:** While Γ is non-empty

- (a) Policy evaluation: Solve

$$\mathbf{v} = \mathbf{r}_\phi + \lambda P_\phi \mathbf{v}.$$

- (b) Policy Improvement: Set $\phi': \mathcal{S} \rightarrow \mathcal{A}$ as

$$\phi'(s) \in \arg \max_{a \in \mathcal{A}} r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s' | s, a) v(s'), \quad s \in \mathcal{S},$$

choosing $\phi'(s) = \phi(s)$ if $\phi(s)$ is an element of $\arg \max$.

- (c) $\Gamma \leftarrow \{s \in \mathcal{S} \mid \phi'(s) \neq \phi(s)\}$
- (d) $\phi \leftarrow \phi'$

3. **Terminate:** Return ϕ and $\mathbf{v}$.

Question 1 (2+2+2 Marks)

Consider an infinite-horizon MDP with discount factor $\lambda < 1$, one-step expected rewards $r(s, a)$ and S - A - S transition matrix $P(s' | s, a)$. Define the quantities

$$r_{\min} = \min_{s \in \mathcal{S}} \min_{a \in \mathcal{A}} r(s, a), \quad r_{\min \max} = \min_{s \in \mathcal{S}} \max_{a \in \mathcal{A}} r(s, a).$$

(a) Prove that the operator $L : \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ is monotone, i.e., if $\mathbf{u} \leq \mathbf{v}$, then $L\mathbf{u} \leq L\mathbf{v}$.

(b) Prove that

$$v(s) = \frac{1}{1 - \lambda} r_{\min} \quad \forall s \in \mathcal{S}$$

satisfies $L\mathbf{v} \geq \mathbf{v}$.

(c) Prove that

$$u(s) = \frac{1}{1 - \lambda} r_{\min \max} \quad \forall s \in \mathcal{S}$$

satisfies $L\mathbf{u} \geq \mathbf{u}$.

The idea here is that if the value iteration algorithm is initialized with $\mathbf{v}_0 \in \{\mathbf{v}, \mathbf{u}\}$, then the sequence of iterates $\{\mathbf{v}_k\}_{k \geq 0}$ arising from the value iteration algorithm converge monotonically to $\mathbf{v}_{\infty, \lambda}^*$, i.e.,

$$\mathbf{v}_0 \leq \mathbf{v}_1 \leq \dots \leq \mathbf{v}_{\infty, \lambda}^*.$$

In this sense, $\mathbf{v}, \mathbf{u}$ as defined above are “good” choices for the initial value.

Question 2 (1+1+2 Marks)

Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as

$$T(x, y) = \begin{bmatrix} 0.8 & 0 \\ 0.8 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

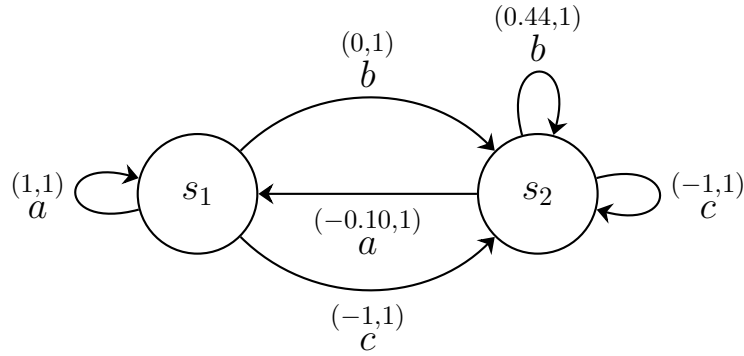
- (a) Prove that T is a contraction in the $\|\cdot\|_\infty$ norm.
- (b) Prove that T is NOT a contraction in the $\|\cdot\|_1$ norm.
- (c) Given $\mathbf{x}_0 \in \mathbb{R}^2$, let $\{\mathbf{x}_k\}_{k \geq 1}$ be defined as $\mathbf{x}_k = T\mathbf{x}_{k-1}$ for all $k \in \mathbb{N}$.
What do the above results say about the convergence of $\{\mathbf{x}_k\}_{k \geq 0}$ in the $\|\cdot\|_\infty$ and $\|\cdot\|_1$ norms?

Question 3 (3+1+1 Marks)

Consider the following infinite-horizon discounted Markov Decision Process (MDP):

$$\mathcal{S} = \{s_1, s_2\}, \quad \mathcal{A} = \{a, b, c\}, \quad \lambda = 0.5.$$

All three actions are available at both states. Each arrow is labeled (r, p) , where r is the immediate reward and p is the transition probability. All transitions are deterministic ($p = 1$).



Starting from $\mathbf{v}_0 = \mathbf{0}$ and $\varepsilon = 0.5$, perform value iteration until the stopping criterion is satisfied. Let $\mathbf{v}_{\infty, \lambda, \varepsilon}$ and ϕ_ε denote respectively the value and the decision rule returned by the value iteration algorithm upon termination.

- Compute the sequence of value iterates $(\mathbf{v}_1, \mathbf{v}_2, \dots)$ until the stopping criterion is satisfied.
- Determine ϕ_ε .
- Compute the true value function $\mathbf{v}_{\infty, \lambda}^{\phi_\varepsilon}$ of the policy $\pi = \{\phi_\varepsilon, \phi_\varepsilon, \dots\}$.